Convergent Multiagent Formation Control With Collision Avoidance

Jinwen Hu, Houxin Zhang, Lu Liu, Xiaoping Zhu, Chunhui Zhao, and Quan Pan

Abstract—**A key problem in the formation control of homogeneous multiagent systems is the collision-free convergence of the agent positions into a desired formation. It is a typical NP-hard problem by considering the problem as optimizing the assignment of multiple destinations to the same number of agents deployed in an open space. It becomes even harder if the collision avoidance is required during the motion of agents, and thus, a suboptimal but efficient solution is adequate. The traditional methods make it by accurate preplanning of the motion trajectory of each single agent, or simply letting them reach an equilibrium as a tradeoff between the collision avoidance and the desired formation. In this article, a distributed control algorithm embedded with an assignment switch scheme is proposed to guarantee that the asymptotic convergence to the desired formation is achieved with no collisions between agents. By the proposed algorithm, the agents keep moving in straight lines toward their respective destinations until they are going to collide if they do not stop, at which moment the agents will communicate their information locally to switch their destination assignments so that they will continue to move in different directions and avoid potential collisions. Distributed control rules are also defined to confine the motion space of each agent for collision avoidance. It has been rigorously proven that the positions of all agents converge to the desired formation with no collision under random initial deployment. In addition, a detailed parameter design procedure is provided for both setting and controlling of the formation. Finally, Monte Carlo simulations and actual experiments in the outdoor environment are implemented and the results verify the effectiveness of the proposed algorithm.**

Index Terms—**Collision avoidance, distributed control, formation control, multiagent systems (MASs).**

I. Introduction

FORMATION control of multiagent systems (MASs) has gained considerable attention in recent years due to its broad potential applications such as search and rescue missions, transportation and logistics, cooperative robotics, and surveillance [1]–[3]. A wide range of topics have been researched including formation keeping/reconfiguration, obstacle avoidance, coverage control, distributed path planning, and cooperative information processing [2], [4], [5]. In large-scale MASs, a fundamental formation control problem is how to let the agents achieve a desired formation distributively without collision between agents.

Convergence to the desired formation by distributed methods has been widely studied in consensus-based methods. In [6], a consensus protocol is applied to deal with the formation control problems of second-order swarm systems. In [7], the leaderless consensus and consensus regulation problems for both first-order and second-order integrators are discussed. A fast dynamic consensus algorithm for a wireless sensor network is proposed in [8] for network formation control. In addition to the asymptotic convergence, finite-time consensus problems have been discussed and many results have been obtained [9]–[13]. With the development of the consensus theory, collision avoidance has been added as a key performance index of formation control of MASs. In [14], consensus protocols are combined with artificial potential fields to guarantee stability of the control scheme while avoiding collisions. The method is further validated by experiments in [15]. In [16], the collision avoidance is considered in the formation control of multi-unmanned-aerial-vehicle (UAV) systems by fusing consensus protocols with model predictive control. In [14], a convergent formation control algorithm is designed with avoidance abilities under the assumption that UAV only take evasive actions on the vertical plane. In [17], a formation control strategy with a consensus-based position estimator is proposed with the collision avoidance using potential field. Similar idea of using the potential field for collision avoidance can also be found in [18]–[21]. Due to the consideration of collision avoidance, it is much difficult to prove the exact convergence to the desired formation as usual by the consensus-based algorithms.

In addition to the consensus-based algorithms, there have been many other methods dealing with the collision-free formation control. In [22], a distributed collision-free formation control method is proposed based on model predictive control (MPC). However, as many other MPC-based formation control methods in MASs [23]–[25], it always requires to solve a high-dimensional nonconvex nonlinear optimization problem. Even though the computation load is affordable, the solution feasibility is difficult to be satisfied in a dense network. In [26], a

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target tracking and formation control method is proposed to let a team of agents track a target by moving as a formation. However, the formation is not required to keep any specific geometric pattern. In [27], an artificial potential field is generated between each free robot and the virtual attractive point of the swarm, under which the free robots will be led to their swarm. It is further extended into the dynamic and noisy environment in [28]. In [29], the artificial bee colony algorithm is employed to solve constrained optimization problems with collision avoidance constraints. In [30], a rapidly exploring random tree algorithm is proposed for collision-free path planning in MASs. In [31], the similar problem of collision-free formation control is considered. However, it only proves the collision avoidance in the case that the paths of two agents intersect with each other if their destinations do not switch, while it neglects other critical cases in which the agents may collide. Moreover, the impact of the motion control on collision avoidance is not considered. These two reasons results in that the proposed method is unreliable for real implementations. In [32], the formation control of MASs is formulated as a task assignment problem where the destinations are assigned to agents distinctively. However, the motion control method adopted from [33] assumes that each agent knows the positions of all other agents at each time instant, which is unrealistic in the distributed systems. In [34], a distributed control strategy is proposed for collision-free formation control. However, it only proves that the formation converges to a local optimum configuration that may not be the desired formation. In [35] and [36], a distributed reciprocal collision avoidance method is proposed by formulating the collision avoidance as a linear programming problem and applied in the formation control of MASs. However, the resulted formation may be trapped in a local equilibrium. In [37], a concurrent assignment and planning of trajectories (CAPT) algorithm is proposed to let MASs achieve any desired formations without collisions. However, the collision-free formation convergence is rigorously proven only in the centralized implementations. Based on the CAPT algorithm and the reciprocal collision avoidance method, smooth trajectories are built using B-splines with downwash effect considered in [38]. However, rigorous proof on the formation convergence is still not provided.

Reinforcement learning has also applied into the motion control of robots with collision avoidance. In [39], a multi-Q learning method is applied to multirobot systems. In [40], reinforcement learning is combined with flocking control to enable multirobot networks to learn how to avoid predators. Though these methods can be applied in more general cases with collision avoidance against different types of obstacles, it is very difficult to provide rigorous proof of the convergence to the desired formation.

This article addresses a fundamental problem in the formation control of homogeneous MASs, i.e., to achieve the desired formation with no collision between agents in a distributed manner under limited communication range and motion capability. Different from the existing methods that only prove the convergence to the desired formation with bounded error or under strict assumptions on the initial conditions, we prove the zero-error formation convergence with collision avoidance by the proposed algorithm, which imposes no restrictions on the initial agent positions except the separation distance. Most importantly, a detailed parameter design is provided so that users can accurately control the minimum separation distance between any two agents while achieving the desired formation.

To do so, the formation control problem is first formulated as a general real-time optimization problem where N destinations need to be assigned to the N agents and the way points for agents moving need to be calculated, which is typically NP hard. Second, a distributed formation control algorithm is proposed to realize efficient real-time computation of suboptimal solutions by introducing the priority number of destinations, which is similar to the common idea of using sequential implementation to solve NP-hard problems. However, our algorithm is implemented by each agent in parallel and distributively, where each agent determines whether to switch its destination with its neighbor and where to move in real time only based on the local information. Third, a detailed parameter design procedure is given for the convenience of users to strictly control the minimum separation distance, moving speed of agents, communication range, initial settings and the design of desired formations, etc. It is proven that by the proposed algorithm and the designed parameters, the desired formation can be achieved asymptotically, where the initial positions of agents can be randomly set except that the minimum separation distance should be satisfied. Finally, Monte Carlo simulations are implemented to testify the effectiveness of the proposed algorithm with random initial settings, which show that the collision avoidance is always guaranteed and the desired formation can be achieved exponentially fast.

II. Problem Formulation

Denote the set of N homogeneous agents by $\mathcal{V} = \{1, 2, \ldots, N\}$ which have the same communication and moving capabilities such as the communication range R_{C} and the maximum moving speed U. Without loss of generality, we use the way-point dynamic model for each agent:

$$p_{i,k+1} = p_{i,k} + v_{i,k}, \tag{1}$$

where k is the discrete-time index, $p_{i,k} \in \mathbb{R}^n$ ($n = 2$ or 3) denotes the position of agent i at time k, $v_{i,k}$ (subject to $\|v_{i,k}\| \leqslant U$) denotes the displacement of agent i during the k-th time interval. Each time interval is split into two subintervals, one of which is taken up by the agent communications for information sharing and the other by the agent moving towards the desired way points. Two agents i and j have a communication link at time k if and only if $\|p_{i,k} - p_{j,k}\| \leqslant R_{\mathrm{C}}$. Therefore, the topology of the agents can be denoted by an unordered graph $\mathcal{G} = \{\mathcal{E}, \mathcal{V}\}$, where $\mathcal{E}$ is the edge set defined as $\mathcal{E}_k = \{\{i,j\} | \|p_{i,k} - p_{j,k}\| \leqslant R_{\mathrm{C}}, j \neq i$ and $i, j \in \mathcal{V}\}$. The neighbor set of the agent i is defined as $\mathcal{N}_{i,k} = \{j | \{i,j\} \in \mathcal{E}_k, j \in \mathcal{V}\}$. Throughout this article, we use $\| \bullet \|$ to denote Euclidean norm, and "T" to denote transpose operation. All the vectors in this article are defined as column vectors by default.

For the clarification of expression and without loss of generality, we schedule fixed time interval for a periodic discrete-time implementation, which includes two fixed consecutive

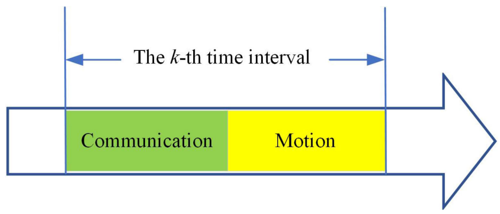


Fig. 1. Determination of time intervals.

subintervals, one for communication and one for motion, as shown in Fig. 1, where the start of each time interval (also the end of its previous time interval) is defined as the time instant at which all agents have completed communications and are ready to move. Thus, during the kth time interval, the time index of all the state variables are denoted by k.

The desired formation of the agents is defined by a set of destinations $\mathcal{Q} = \{q_1, q_2, \ldots, q_N\}$ ($q_m \neq q_n$ for $m \neq n, m, n \in \mathcal{V}$). The formation is said to be achieved if each destination in $\mathcal{Q}$ is occupied by one agent in $\mathcal{V}$. Obviously, this implies a one to one mapping $\phi : \mathcal{V} \to \mathcal{Q}$ that assigns a destination to each agent. ϕ is allowed to change according to some rules, and thus, the destination assignment of the agent i at time k is denoted by $\phi_{i,k} \in \mathcal{Q}$. Denoting the group destination assignment vector by $\phi_k = [\phi_{1,k}, \phi_{2,k}, \ldots, \phi_{N,k}]$, we can see that there are totally $N!$ possible assignments at each time, the set of which is defined as $\mathcal{A}$. Denote the group position vector by $p_k = [p_{1,k}^{\mathrm{T}}, p_{2,k}^{\mathrm{T}}, \ldots, p_{N,k}^{\mathrm{T}}]^{\mathrm{T}}$ and the group motion vector by $v_k = [v_{1,k}^{\mathrm{T}}, v_{2,k}^{\mathrm{T}}, \ldots, v_{N,k}^{\mathrm{T}}]^{\mathrm{T}}$, then we can define the allowable motion set for agents moving as $\mathcal{U} = \{v_k | \|v_{i,k}\| \leqslant U \ \ \forall i \in \mathcal{V}\}$. Based on the aforementioned definitions, the formation control problem is formulated as the following optimization problem:

$$\begin{aligned} &\min_{v_k \in \mathcal{U}, \phi_k \in \mathcal{Q}} \max_{i \in \mathcal{V}} \|p_{i,k} + v_{i,k} - \phi_{i,k}\|, \\ &\text{s.t.} \ \ p_{i,k} \neq p_{j,k} \ \ \forall i, j \in \mathcal{V} \ \ \text{and} \ \ i \neq j \end{aligned} \tag{2}$$

where v_k and ϕ_k are the variables to be controlled for optimization. The constraint $p_{i,k} \neq p_{j,k}$ is imposed to avoid collision between agents.

The optimization problem (2) is a typical NP-hard problem. Therefore, our aim is to achieve the desired formation with suboptimal solutions subject to the same constraints as in (2), where the computation is implemented distributively for each agent. In Section III, we will propose a distributed control strategy to make the agents converge to their respective assigned destinations, i.e., to let $\|p_{i,k} - \phi_{i,k}\|$ converge to zero, with no collision between agents.

III. Partition-Based Distributed Formation Control With Adaptive Assignment Switch

The traditional methods to avoid collision for mobile agents is to add an expelling force between the neighboring agents, which has a risk of being trapped into a local minimum configuration where the expelling force equals the negative potential force from other sources for each agent [41]. In this article, the collision avoidance is achieved by limiting the movement of each agent to be within its own task region, which is obtained via a partition over the whole space. However, this does not help to achieve the desired formation for the whole team if one agent needs to pass through the task region of another to reach its assigned destination. In such a case, the location assignment has to be switched to a more appropriate one so that each individual agent can keep converging while staying in its task region.

A. Partition-Based Control for Collision Avoidance

In light of this motivation, we first define the whole admissible space for team movement as a convex set $\mathbb{S} \subset \mathbb{R}^n$ that contains all the destinations in $\mathcal{Q}$ and initial assignments in ϕ_0, and also define for $i, j \in \mathcal{V}$

$$C_{i,j,k} = \{s \in \mathbb{S} | \, \|s - p_{i,k}\| \leqslant \|s - p_{j,k}\|\}. \tag{3}$$

We further define

$$S_{i,j,k} = \begin{cases} C_{i,j,k} \backslash \left(C_{i,j,k} \bigcap C_{j,i,k}\right), & \text{if } j < i \\ C_{i,j,k} & \text{otherwise} \end{cases} \tag{4}$$

and

$$S_{i,k} = \bigcap_{j=1}^{N} S_{i,j,k}. \tag{5}$$

In (4), $C_{i,j,k} \bigcap C_{j,i,k}$ stands for the common boundary of $C_{i,j,k}$ and $C_{j,i,k}$, where $\|s - p_{i,k}\| = \|s - p_{j,k}\|$ for $s \in C_{i,j,k} \bigcap C_{j,i,k}$. Such a definition is to make $S_{i,j,k} \bigcap S_{j,i,k} = \emptyset$ and $S_{i,j,k} \bigcup S_{j,i,k} = \mathbb{S}$, which implies that $S_{i,k} \bigcap S_{j,k} = \emptyset$ and $\bigcup_{i=1}^{N} S_{i,k} = \mathbb{S}$. $\{S_{1,k}, S_{2,k}, \ldots, S_{N,k}\}$ is the so-called Voronoi partition of $\mathbb{S}$ and $S_{i,k}$ is the so-called Voronoi cell of the agent i, which encloses $p_{i,k}$ [42].

Note that the computation of $S_{i,k}$ relies on the information of $S_{i,j,k}$ from all $j \in \mathcal{V}$, which is difficult to be obtained in a distributed network with finite communication range. From the view of each agent i, a local Voronoi cell that only requires the local information can be defined as

$$S_{i,k}^{*} = \bigcap_{j \in \mathcal{N}_{i,k}} S_{i,j,k}. \tag{6}$$

Following (6), it can only be implied that $S_{i,k}^{*} \bigcap S_{j,k}^{*} = \emptyset$ for $j \in \mathcal{N}_{i,k}$, i.e., there may be overlaps between $S_{i,k}^{*}$ and $S_{j,k}^{*}$ for some $j \notin \mathcal{N}_{i,k}$ and $j \neq i$. However, if two agents i and j are going to collide, i.e., $\|p_{i,k} - p_{j,k}\| < \delta$ for some small positive number $\delta < R_{\mathrm{C}}$, they should have been the neighbors of each other, which implies that $S_{i,k}^{*} \bigcap S_{j,k}^{*} = \emptyset$. Therefore, $S_{i,k}^{*}$ can be taken as the *Task Region* of the agent i, the computation of which only relies on the local information. Note that each agent i can only compute its own task region $S_{i,k}^{*}$, but can compute $S_{i,j,k}$ for $j \in \mathcal{N}_{i,k}$.

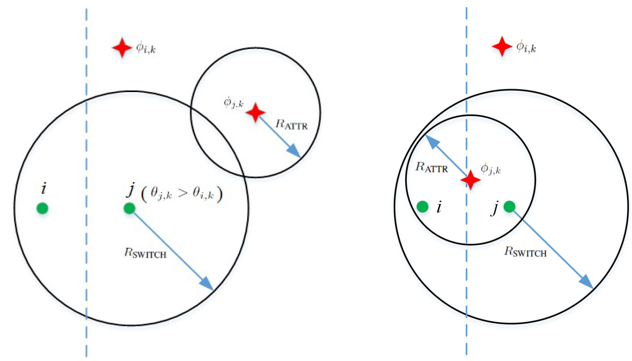
Based on the aforementioned analysis, it is straightforward to get the following result.

Lemma 1: The collision between any agents can be avoided if $v_{i,k}$ is selected such that $p_{i,k+1} \in S_{i,k}^{*}$ is satisfied for all $i \in \mathcal{V}$ and $k > 0$, i.e., each agent moves only inside its own task region.

Thus, we can design the following motion control rule.

Rule 1: Set $v_{i,k} = 0$ if there exists an agent $j \in \mathcal{N}_{i,k}$ such that $\phi_{i,k} \in S_{j,i,k}$ and $\|p_{i,k} - p_{j,k}\| \leqslant R_{\text{STOP}}$, where R_{STOP} is a predefined threshold for collision avoidance.

Remark 1: Rule 1 can guarantee that as long as an agent has a possibility to walk into the task region of a neighbor while they are too close, i.e., being possible to collide with a neighbor, the agent will stop moving. At this moment, one would think to find a path for the agent i to go around the neighbors. However, it is another NP hard problem to find the optimal paths for all agents, where the movement of each agent affects the movement of another. An effective way to let them continue moving is to switch the destination assignments between the two agents or with others so that they change their moving directions. Therefore, in the following part, we will establish the rules for the assignment switch so that the desired formation can still be achieved when such stopping happens.



(a) (b)

Fig. 2. Examples of two types of blockers of the agent i (the dash line is the boundary between $S_{i,j,k}$ and $S_{j,i,k}$). (a) $j \in B^{+}_{i,k}$. (b) $j \in B^{++}_{i,k}$.

B. Distributed Assignment Switch

The basic communication protocol during each communication subinterval is stated as follows.

1) Each agent broadcasts the state information to its neighbors.
2) After receiving the information from all neighbors, an agent determines whether it needs to switch its destination with a neighbor, and if it needs, a switch request is sent to the desired neighbor.
3) If an agent receives any switch requests, it has to choose one of the requesters to switch their destination assignments. When it chooses a requester, it updates its destination to be the one of the chosen requester, and then, sends a reply message to it.
4) After receiving the reply message from its neighbor, an agent automatically updates its destination to be the one of that neighbor.

There are three basic issues to consider for the assignment switch of an agent, which are as follows.

1) Whether it needs a switch?
2) To whom a switch request should be sent?
3) Whose request should be agreed if multiple requests are received?

The main challenge in solving the aforementioned issues is that the decisions must be made distributively due to the limited communication range, which may result in conflicting decisions of neighbors instead of one common optimal decision as obtained in the centralized optimization.

To solve these issues, we first introduce a so-called priority number associated with each destination $\phi_{i,k}$ as

$$\theta_{i,k} \in \mathcal{V} = \{1, 2, \ldots, N\} \tag{7}$$

where $\theta_{i,k} \neq \theta_{j,k}$, for $i \neq j$. Thus, θ is actually a mapping from $\mathcal{V}$ to $\mathcal{V}$ at a given time k, where the former one is the agent set and the latter one is the priority number set. The smaller $\theta_{i,k}$ is, the higher the priority of the destination $\phi_{i,k}$ is. Without loss of generality, the index number of each destination in $\mathcal{Q}$ can be defined to be the same as the priority number, i.e., the priority number of destination q_m is m $(m \in \mathcal{V})$. The priority number of each destination can be arbitrarily defined at the initial time. The main reason of using the priority number is to solve the NP-hard problem (2) in a sequential manner as will be shown later so that a feasible solution can be found efficiently.

Define the set of neighbors that block the agent i as

$$B_{i,k} = \{j \in \mathcal{N}_{i,k} | \phi_{i,k} \in S_{j,i,k},\ \|p_{i,k} - p_{j,k}\| \leqslant R_{\text{SWITCH}}\} \tag{8}$$

where R_{SWITCH} is a predefined range for the assignment switch. Each agent $j \in B_{i,k}$ is called as a "blocker" of the agent i. Further, define the following two specific types of blockers, which are the subsets of $B_{i,k}$:

$$B^{+}_{i,k} = B_{i,k} \cap \{j \in \mathcal{N}_{i,k} | \theta_{j,k} > \theta_{i,k},\ p_{j,k} \notin A_{j,k}\}$$
$$B^{++}_{i,k} = B_{i,k} \cap \{j \in \mathcal{N}_{i,k} | p_{i,k} \in A_{j,k},\ p_{j,k} \in A_{j,k}\} \tag{9}$$

where $A_{i,k}$ is the attractive neighborhood centered at destination $\phi_{i,k}$ with a predefined radius R_{ATTR} defined as

$$A_{i,k} = \{s \in \mathbb{S} |\ \|s - \phi_{i,k}\| < R_{\text{ATTR}}\}. \tag{10}$$

$B^{+}_{i,k}$ consists of the blockers that are located outside their respective attractive neighborhoods but have larger priority numbers than the agent i has. $B^{++}_{i,k}$ consists of the blockers that are located within their respective attractive neighborhoods that also enclose the agent i. A geometric illustration of the two types of blockers is shown in Fig. 2.

The aforementioned ranges can be always designed such that $R_{\text{C}} \geqslant R_{\text{SWITCH}} \geqslant 2R_{\text{STOP}}, R_{\text{ATTR}} \geqslant 2R_{\text{STOP}}$. Moreover, we can get $A_{i,0} \cap A_{j,0} = \emptyset$ by letting R_{ATTR} satisfy

$$d^* = \min_{m,n \in \mathcal{V}, m \neq n} \|q_m - q_n\| \geqslant 2R_{\text{ATTR}} \tag{11}$$

where d^* determines how close the neighboring agents should be when they are in the desired formation.

Remark 2: Bearing in mind that Q is time invariant and $\phi_{i,k} \in Q$ for all $i \in \mathcal{V}$, we can get that $A_{i,0} \cap A_{j,0} = \emptyset$ implies $A_{i,k} \cap A_{j,k} = \emptyset$ for all $k > 0$. Note that $A_{i,k} \cap A_{j,k} = \emptyset$ implies that $|B^{++}_{i,k}| \leqslant 1$, i.e., one agent can only be inside at most one attractive neighborhood.

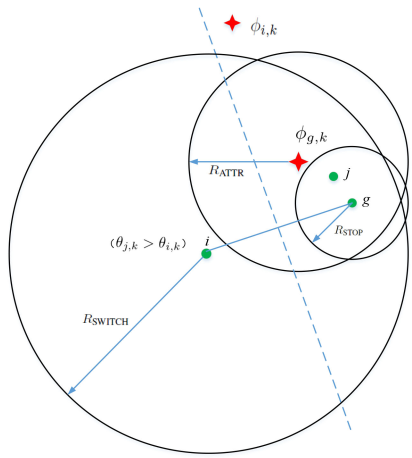


Fig. 3. Example of the agent $j \in G_{i,k}$ (the dash line is the boundary between $S_{i,g,k}$ and $S_{g,i,k}$).

Similar to d^*, we can define the actual minimum distance between agents as

$$d_k = \min_{i,j \in \mathcal{V}, i \neq j} \|p_{i,k} - p_{j,k}\| . \tag{12}$$

The initial minimum distance d_0 should satisfy some condition for collision avoidance and will be discussed more in detail in the following part.

Define the following subset of neighbors of the agent i:

$$\begin{aligned} D_{i,k} = &\{j \in \mathcal{N}_{i,k} | \theta_{j,k} < \theta_{i,k},\ p_{j,k} \notin A_{j,k},\ i \in B_{j,k}\} \\ &\bigcup \{j \in \mathcal{N}_{i,k} | i \in G_{j,k}\} \end{aligned} \tag{13}$$

where $G_{i,k}$ illustrated by Fig. 3 is defined as

$$\begin{aligned} G_{i,k} = \Big\{ &j \in \mathcal{N}_{i,k} | \theta_{j,k} > \theta_{i,k},\ \|p_{i,k} - p_{j,k}\| \leqslant R_{\text{SWITCH}} \\ &\|p_{j,k} - p_{g,k}\| \leqslant R_{\text{STOP}}, j \in B_{g,k} \\ &g \in B_{i,k},\ p_{g,k} \in A_{g,k} \Big\}. \end{aligned} \tag{14}$$

Then, we can design the following assignment switch rules.

Rule 2: If $p_{i,k} \notin A_{i,k}$, $D_{i,k} = \emptyset$ and $B^{+}_{i,k} \cup B^{++}_{i,k} \neq \emptyset$, then the agent i sends a switch request to the agent j^*, where

$$j^* = \underset{j \in B^{+}_{i,k} \cup B^{++}_{i,k}}{\arg\min} \|\phi_{i,k} - p_{j,k}\| . \tag{15}$$

If more than one agent give the minimum of $\|\phi_{i,k} - p_{j,k}\|$, j^* is the smallest of them, i.e., $\phi_{i,k} \in \cup_{j \in B^{+}_{i,k} \cup B^{++}_{i,k}} S_{j^*,j,k}$.

Rule 3: If $p_{i,k} \notin A_{i,k}$, $D_{i,k} = \emptyset$, $B^{+}_{i,k} \cup B^{++}_{i,k} = \emptyset$, and $G_{i,k} \neq \emptyset$, then the agent i sends a switch request to the agent j^*, where

$$j^* = \underset{j \in G_{i,k}}{\arg\min} \|\phi_{i,k} - p_{j,k}\| . \tag{16}$$

If more than one agent give the minimum of $\|\phi_{i,k} - p_{j,k}\|$, j^* is the smallest of them, i.e., $\phi_{i,k} \in \cup_{j \in G_{i,k}} S_{j^*,j,k}$.

Rule 4: If the agent i receives switch requests from one or more neighbors, then it selects the neighbor j with the smallest priority number for assignment switch and sets $\phi_{i,k+1} = \phi_{j,k}$, $\theta_{i,k+1} = \theta_{j,k}$. After that, it sends a reply to the agent j. When the agent j receives the reply from the agent i, it sets $\phi_{j,k+1} = \phi_{i,k}$, $\theta_{j,k+1} = \theta_{i,k}$.

Rule 5: In all other cases where any one of the conditions listed in Rules 2 and 3 does not hold for the agent i, it does not send any switch request.

Remark 3: Rule 2 to Rule 5 tell that if an agent decides to send a switch request, it will not receive a switch request from any neighbor at the same time. Conversely, if an agent receives a switch request from any neighbor, it should not have sent a switch request at the same time. Such design is to avoid the case, e.g., the agent i sends a request to the agent j and the agent j sends a request to l at the same time, in which the decision of the agent j on whether to switch with the agent i relies on the decision of l, and thus, their decisions cannot be carried out in parallel and distributively.

C. Motion Control

According to Rule 1, an agent may be forced to stop moving if specific conditions are satisfied. When it is allowed to move, an appropriate motion control scheme has to be designed. Given the destination $\phi_{i,k}$, the most straightforward motion control for the agent i at time k is to let it walk toward $\phi_{i,k}$ in a straight line, i.e., to set

$$v_{i,k} = \lambda_{i,k} (\phi_{i,k} - p_{i,k}) \tag{17}$$

where $0 \leqslant \lambda_{i,k} \leqslant 1$ is a scaling factor. If the conditions of Rule 1 do not hold, we let

$$\lambda_{i,k} = \Gamma (\|\phi_{i,k} - p_{i,k}\| , Z_{i,k}) \tag{18}$$

where $Z_{i,k}$ is a positive number to control the step length and $\Gamma(a,\ b)$ $(a > 0,\ b > 0)$ is a saturation function defined as

$$\Gamma(a,b) = \begin{cases} 1, & \text{if } a \leqslant b \\ \frac{b}{a}, & \text{otherwise.} \end{cases} \tag{19}$$

Based on (17), we design the following motion control rule.

Rule 6: Set $Z_{i,k} = \mu U$ in (18) if there exists an agent $j \in \mathcal{F}_{i,k}$ such that $\|p_{i,k} - p_{j,k}\| = \min_{g \in \mathcal{N}_{i,k}} \|p_{i,k} - p_{g,k}\|$, where $0 < \mu \leqslant 1$ is a scaling parameter and

$$\begin{aligned} \mathcal{F}_{i,k} = \big\{ &j \in \mathcal{N}_{i,k} | R_{\text{STOP}} < \|p_{i,k} - p_{j,k}\| \leqslant R_{\text{STOP}} + 2U \\ &(\phi_{i,k} - p_{i,k})^{\mathrm{T}} (p_{j,k} - p_{i,k}) > 0 \big\}. \end{aligned} \tag{20}$$

Otherwise, set $Z_{i,k} = U$, where $U > 0$ is the maximum step length.

Remark 4: Setting $\mu < 1$ is actually to reduce the moving speed of the agents when they satisfy the conditions in Rule 6. $R_{\text{STOP}} < \|p_{i,k} - p_{j,k}\| \leqslant R_{\text{STOP}} + 2U$ can be seen as a buffer zone, where the agents may need to slow down. In fact, more than one buffer zones sequentially connected can be set to give smoother change of speeds if the designed μ is too small. If H buffer zones are set, the hth buffer zone is defined as $R_{\text{STOP}} + 2\mu_{h-1} U < \|p_{i,k} - p_{j,k}\| \leqslant R_{\text{STOP}} + 2\mu_h U$ $(h = 1, 2, \ldots, H)$ and its associated scaling factor μ_h should be subject to $0 = \mu_0 < \mu_h < \mu_{h+1} \leqslant 1$. If $\|p_{i,k} - p_{j*,k}\|$ is located in the hth buffer zone where $j* = \arg\min_{j \in \mathcal{F}_{i,k}} \|p_{i,k} - p_{j*,k}\|$, set $Z_{i,k} = \mu_h U$. Note that $Z_{i,k} = U$ if $\min_{j \in \mathcal{N}_{i,k}} \|p_{i,k} - p_{j,k}\| \leqslant R_{\text{STOP}}$ and $\phi_{i,k} \in S^*_{i,k}$ hold. This property will be used in Section III-E.

D. Formation Control Algorithm and Convergence Analysis

Now, we integrate the aforementioned assignment switch and motion control rules into the distributed formation control algorithm as depicted in Algorithm 1.

Before giving the main results on the formation convergence, we define some auxiliary variables. Define the inverse mapping of θ as

$$\vartheta : \mathcal{V} \to \mathcal{V} \tag{21}$$

such that $\vartheta_{m,k}$ is the agent that takes q_m as its destination, where $\theta_{\vartheta_{m,k},k} = m$ and $\phi_{\vartheta_{m,k},k} = q_m$.

Moreover, in Algorithm 1, some variables are updated ahead of the time index update before and after the communication (or the motion correspondingly), which we need to compare for the purpose of convergence analysis. In fact, the variables updated within the kth time interval are exactly the same as those before the update within the $(k+1)$th time interval. Therefore, to avoid confusion of notations for such variables, we clarify that each variable with the subscript k denotes its state before the update within the kth time interval. For examples, $\vartheta_{m,k+1}$ denotes either the agent taking q_m as the destination before the communication update within the $(k+1)$th time interval or the agent taking q_m as the destination after the communication update within the kth time interval; $p_{i,k+1}$ denotes either the position of the agent i before the motion update within the $(k+1)$th time interval or the position of the agent i after the motion update within the kth time interval.

We first show that the motion control law and rules never lead to an increase of the distance from an agent to its destination by the following lemma.

Lemma 2: For all $m \in \mathcal{V}$ and $k \geqslant 0$, if $q_m \neq p_{\vartheta_{m,k+1},k}$, then $\|q_m - p_{\vartheta_{m,k+1},k+1}\| \leqslant \|q_m - p_{\vartheta_{m,k+1},k}\|$ holds and the equality holds if and only if there exists an agent $j \in \mathcal{N}_{\vartheta_{m,k+1},k}$ such that $q_m \in S_{j,\vartheta_{m,k+1},k}$ and $\|p_{\vartheta_{m,k+1},k} - p_{j,k}\| \leqslant R_{\text{STOP}}$.

Proof: The conclusion directly follows Rules 1 and 6 due to the straight line movement of the agent $\vartheta_{m,k+1}$ toward q_m within the kth time interval. ■

For an agent taking q_m as its destination, if it gets its destination assignment switched by sending a switch request to its neighbor, we call such switch as an active switch for q_m; if it does so by receiving a switch request from its neighbor, we call such switch as an passive switch for q_m. Denote by $\tau_{m,\kappa}$ the time of the κth active switch for q_m, then we can show that there must be a decrease of the distance between q_m and $p_{\vartheta_{m,k}}$ by at most $N+2$ active switches in the following lemma.

Lemma 3: If there is no passive switch for q_m $(m \in \mathcal{V})$ during time range $[\tau_{m,\kappa}, \bar{\tau}]$ subject to $\kappa \geqslant 1$ and $\bar{\tau} \geqslant \tau_{m,\kappa+2}$, then it holds that $\omega_{m,\kappa+1} \leqslant \omega_{m,\kappa}$, where

$$\begin{aligned} \omega_{m,\kappa} &= \min\left\{\xi_{m,\tau_{m,\kappa}},\ \xi_{m,\tau_{m,\kappa+1}}\right\} \\ \xi_{m,k} &= \left\|q_m - p_{\vartheta_{m,k},k}\right\|. \end{aligned} \tag{22}$$

Algorithm 1: Distributed Formation Control With Collision Avoidance.

Initialize the parameters subject to $k = 0$, $\theta_{i,0} \in \mathcal{V}$, $\phi_{i,0} \in \mathcal{Q}$, $0 < \mu \leqslant 1$, $R_{\text{STOP}} > 2U$, $R_{\text{C}} \geqslant R_{\text{SWITCH}} \geqslant 2R_{\text{STOP}}$, $R_{\text{ATTR}} \geqslant 2R_{\text{STOP}}$, $d^* \geqslant 2R_{\text{ATTR}}$ and $d_0 > 0$.
for $i = 1$ to N **do**
  Broadcast $p_{i,k}$, $\phi_{i,k}$, $\theta_{i,k}$ to neighbors;
  Receive $p_{j,k}$, $\phi_{j,k}$, $\theta_{j,k}$ from $j \in \mathcal{N}_{i,k}$;
  Compute $D_{i,k}$ following (13);
  if $\|\phi_{i,k} - p_{i,k}\| > R_{\text{ATTR}}$ and $D_{i,k} = \emptyset$ **then**
    Compute $B_{i,k}$, $B^+_{i,k}$, $B^{++}_{i,k}$ following (8) and (9)
    if $B^+_{i,k} \cup B^{++}_{i,k} \neq \emptyset$ **then**
      Compute j^* following (15);
    else
      Compute $G_{i,k}$ following (14);
      if $G_{i,k} \neq \emptyset$ **then**
        Compute j^* following (14);
      end if
    end if
  end if
  Send a switch request to the agent j^* if it exists;
  if multiple switch requests are received **then**
      Find the requester j with the smallest priority number;
      Update the destination assignment by $\phi_{i,k} \leftarrow \phi_{j,k}$, $\theta_{i,k} \leftarrow \theta_{j,k}$;
      Send a reply message to the agent j;
  end if
  if a reply message is received from $j \in \mathcal{N}_{i,k}$ **then**
    Update the destination assignment by $\phi_{i,k} \leftarrow \phi_{j,k}$, $\theta_{i,k} \leftarrow \theta_{j,k}$;
  end if
  Broadcast the updated $\phi_{i,k}$, $\theta_{i,k}$ to neighbors;
  Receive $\phi_{j,k}$, $\theta_{j,k}$ from $j \in \mathcal{N}_{i,k}$;
  if there exists an agent $j \in \mathcal{N}_{i,k}$ such that $\phi_{i,k} \in S_{j,i,k}$ and $\|p_{i,k} - p_{j,k}\| \leqslant R_{\text{STOP}}$ **then**
    Set $v_{i,k} = 0$;
  else
    Compute $v_{i,k}$ following (17) and move to the new location $p_{i,k} + v_{i,k}$;
  end if
  Update location by $p_{i,k} \leftarrow p_{i,k} + v_{i,k}$;
end for
Update time index by $k \leftarrow k + 1$.

If $\bar{\tau} \geqslant \tau_{m,\kappa+N+2}$, it further holds that $\omega_{m,1} - \omega_{m,N+1} > 0$.

Proof: According to Rules 2 and 3, the agent $\vartheta_{m,k}$ can only send a switch request to a neighbor in $B^+_{\vartheta_{m,k},k} \cup B^{++}_{\vartheta_{m,k},k} \cup G_{\vartheta_{m,k},k}$ and when $\xi_{m,k} > R_{\text{ATTR}}$.

If at time k an assignment switch occurs between agents $\vartheta_{m,k}$ and $\vartheta_{m,k+1} \in B^+_{\vartheta_{m,k},k} \cup B^{++}_{\vartheta_{m,k},k}$, then it holds following Rule 2 and Lemma 2 that

$$\xi_{m,k+1} \leqslant \left\|q_m - p_{\vartheta_{m,k+1},k}\right\| \leqslant \xi_{m,k}. \tag{23}$$

According to the definition of $S_{\vartheta_{m,k+1},\vartheta_{m,k},k}$ in (5), if $\xi_{m,k+1} = \|q_m - p_{\vartheta_{m,k+1},k}\| = \xi_{m,k}$ holds, we have $\vartheta_{m,k+1} < \vartheta_{m,k}$.

If at time k an assignment switch occurs between agents $\vartheta_{m,k}$ and $\vartheta_{m,k+1} \in G_{\vartheta_{m,k},k}$, then there exists an agent g subject to $g \in B_{\vartheta_{m,k},k}$, $\vartheta_{m,k+1} \in B_{g,k}$, $\vartheta_{m,k} \notin A_{g,k}$, $g \in A_{g,k}$ and $\|p_{g,k} - p_{\vartheta_{m,k+1},k}\| \leqslant R_{\text{STOP}}$. According to the Rule 1, the agent g will not move at time k. $g \in B_{\vartheta_{m,k},k}$ implies $\|q_m - p_{g,k}\| \leqslant \|q_m - p_{\vartheta_{m,k},k}\|$ and if the equality holds, $g < \vartheta_{m,k}$ is further implied. Then, we can discuss in the following two cases:

First, if $q_m \in S_{\vartheta_{m,k+1},g,k}$, we can get

$$\xi_{m,k+1} \leqslant \left\|q_m - p_{\vartheta_{m,k+1},k}\right\| \leqslant \|q_m - p_{g,k}\| \leqslant \xi_{m,k} \quad (24)$$

where $\|q_m - p_{\vartheta_{m,k+1},k}\| = \|q_m - p_{g,k}\|$ implies $\vartheta_{m,k+1} < g$. Hence, $\xi_{m,k+1} = \xi_{m,k}$ results in $\vartheta_{m,k+1} < g < \vartheta_{m,k}$.

Second, if $q_m \in S_{g,\vartheta_{m,k+1},k}$, we have

$$\|q_m - p_{g,k}\| \leqslant \left\|q_m - p_{\vartheta_{m,k+1},k}\right\| \quad (25)$$

where $\|q_m - p_{g,k}\| \leqslant \|q_m - p_{\vartheta_{m,k+1},k}\|$ implies $g < \vartheta_{m,k+1}$. In addition, it holds that $g \in B^{++}_{\vartheta_{m,k+1},k+1}$. Then, at time $k+1$, the agent $\vartheta_{m,k+1}$ will send a switch request to the agent $\vartheta_{m,k+2} \in B^{+}_{\vartheta_{m,k+1},k+1} \cup B^{++}_{\vartheta_{m,k+1},k+1}$ following Rule 2 so that:

$$\begin{aligned}\xi_{m,k+2} \leqslant \left\|q_m - p_{\vartheta_{m,k+2},k+1}\right\| &\leqslant \|q_m - p_{g,k+1}\| \\ &= \|q_m - p_{g,k}\| \\ &\leqslant \xi_{m,k}\end{aligned} \quad (26)$$

where $\|q_m - p_{\vartheta_{m,k+2},k+1}\| = \|q_m - p_{\vartheta_{m,k},k}\|$ implies $\vartheta_{m,k+2} \leqslant g$. Thus, $\xi_{m,k+2} = \xi_{m,k}$ implies $\vartheta_{m,k+2} < \vartheta_{m,k}$.

Letting $\tau_{m,\kappa} = k$ and since there is no passive switch between $\tau_{m,\kappa}$ and $\tau_{m,\kappa+1}$, we have $\xi_{m,\tau_{m,\kappa+1}} \leqslant \xi_{m,1+\tau_{m,\kappa}}$ directly following Lemma 2. In sum of all the aforementioned cases, we can get $\omega_{m,\kappa+1} \leqslant \omega_{m,\kappa}$.

Suppose $\omega_{m,1} - \omega_{m,N+1} = 0$, then we get $\omega_{m,\kappa} = \omega_{m,\kappa+1}$ for $1 \leqslant \kappa \leqslant N$. According to the aforementioned derivation, it holds that $\rho_{m,\kappa+1} < \rho_{m,\kappa}$ for $1 \leqslant \kappa \leqslant N$, where

$$\rho_{m,\kappa} = \min\left\{\vartheta_{m,\tau_{m,\kappa}},\ \vartheta_{m,\tau_{m,\kappa+1}}\right\}. \quad (27)$$

Since $\rho_{m,\kappa} \in \mathcal{V}$ for $m \in \mathcal{V}$, we get $\rho_{m,N+1} < \rho_{m,1} - (N-1) \leqslant 1$, which is contradictory to $\rho_{m,N+1} \geqslant 1$. Therefore, $\omega_{m,1} - \omega_{m,N+1} > 0$. ■

The following lemma shows that the convergence of the agents to their respective destinations is equivalent to the convergence into their respective attractive neighborhoods.

Lemma 4: For all $m \in \mathcal{V}$ and $k \geqslant 0$, $\|q_m - p_{\vartheta_{m,k},k}\| < R_{\text{ATTR}}$ implies $\|q_m - p_{\vartheta_{m,t},t}\| < R_{\text{ATTR}}$ for $t > k$.

Proof: According to Rules 2–5, if $\|q_m - p_{\vartheta_{m,k},k}\| < R_{\text{ATTR}}$ (i.e., $p_{\vartheta_{m,k},k} \in A_{\vartheta_{m,k},k}$), the agent $\vartheta_{m,k}$ never sends any switch request to its neighbors and the neighbors located out of $A_{\vartheta_{m,k},k}$ never send any switch request to the agent $\vartheta_{m,k}$ at time k. It can only receive switch requests from the neighbors within $A_{\vartheta_{m,k},k}$. Thus, no matter by communication or motion, $\vartheta_{m,t} \in A_{\vartheta_{m,k},t}$ always holds for $t > k$. ■

Based on the aforementioned conclusions, we can now show the asymptotic convergence of the agents to their respective destinations with collision avoidance. The main idea is to show the convergence of formation in the sequence of the destination priorities.

Theorem 1: By implementing Algorithm 1, the desired formation can be achieved asymptotically without collision between agents, i.e., $\lim_{k\to+\infty} p_{i,k} = \phi_{i,k}$, while $\|p_{i,k} - p_{j,k}\| > 0$ for all $k > 0$, $i, j \in \mathcal{V}$, $i \neq j$.

Proof: First, we prove the conclusion on collision avoidance. Following (17), Rules 1 and 6 with $R_{\text{STOP}} > 2U$, it holds that $p_{i,k} \in S^*_{i,k}$ for all $k > 0$ and $i \in \mathcal{V}$. Then, according to Lemma 1, the collision between any agents is avoided.

Second, we prove the conclusion on formation convergence. Following Remark 2, if $p_{i,k} \in A_{i,k}$ holds for all $i \in \mathcal{V}$, then $p_{i,k}$ will converge to $\phi_{i,k}$ in finite time by straight line movement. This is due to the fact that $A_{i,k} \cap A_{j,k} = \emptyset$. Therefore, we only need to show that $p_{i,k}$ converges to $A_{i,k}$ asymptotically for all $i \in \mathcal{V}$.

We start from $m = 1$ and consider the convergence of agent $\vartheta_{1,k}$ given $\|q_1 - p_{\vartheta_{1,k},k}\| \geqslant R_{\text{ATTR}}$. Since q_1 has the smallest priority number, the agent $\vartheta_{1,k}$ never receives any switch request from neighbors, i.e., there is no passive switch for q_1. Whenever the motion of the agent $\vartheta_{1,k}$ is stopped due to Rule 1 and $B^{+}_{\vartheta_{1,k},k} \cup B^{++}_{\vartheta_{1,k},k} \cup G_{\vartheta_{1,k},k} \neq \emptyset$, it will send a switch request according to Rules 2 and 3. Following Lemma 3, by active switches, $\omega_{m,\kappa}$ will decrease until $\|q_1 - p_{\vartheta_{1,k},k}\| < R_{\text{ATTR}}$.

By excluding the situations in which the agent $\vartheta_{1,k}$ can send a switch request or move as stated in Rules 1–6, it is easy to get the only situation in which the motion of the agent $\vartheta_{1,k}$ is stopped, while no switch request is sent: $B^{+}_{\vartheta_{1,k},k} \cup B^{++}_{\vartheta_{1,k},k} = \emptyset$, $G_{\vartheta_{1,k},k} = \emptyset$, and $B_{\vartheta_{1,k},k}$ only contains a single agent j subject to $\|p_{j,k} - p_{\vartheta_{1,k},k}\| \leqslant R_{\text{STOP}}, p_{\vartheta_{1,k},k} \notin A_{j,k}, p_{j,k} \in A_{j,k}$. In this situation, $G_{\vartheta_{1,k},k} = \emptyset$ implies that the motion of the agent j will not be stopped until it reaches $\phi_{j,k}$ or its motion is stopped. Either of these two cases will occur in finite time according to Rule 6. When either of these two cases occurs at some time $t > k$ and if $j \in B_{\vartheta_{1,t},t}$ and $\|p_{j,t} - p_{\vartheta_{1,t},t}\| \leqslant R_{\text{STOP}}$ still hold, we must have $B^{+}_{\vartheta_{1,k},k} \cup B^{++}_{\vartheta_{1,k},k} \neq \emptyset$ so that there will be an active switch at time t for q_1.

Based on the aforementioned derivation, $\|q_1 - p_{\vartheta_{1,k},k}\|$ can only keep constant for finite length of time and will asymptotically decrease to be smaller than R_{ATTR} as k increases to infinity.

Next, we make a deduction by assuming that $\|q_m - p_{\vartheta_{m,k},k}\| \leqslant R_{\text{ATTR}}$ holds for all $m < L$ at time k, where L is a positive integer smaller than N. Following Lemma 4, $\|q_m - p_{\vartheta_{m,t},t}\| < R_{\text{ATTR}}$ will hold for $t > k$. According to Rules 2 and 3, the agent $\vartheta_{m,k}$ will never send any switch request to neighbors, but still can receive switch requests from neighbors.

We now consider the convergence of the agent $\vartheta_{L,k}$ given $\|q_L - p_{\vartheta_{L,k},k}\| \geqslant R_{\text{ATTR}}$. Since destination q_L has the smallest priority number among the destinations q_m $(L \leqslant m \leqslant N)$, the agent $\vartheta_{L,k}$ never receives any switch request from its neighbors, i.e., there is no passive switch for q_L. Moreover, when agents $\vartheta_{m,k}$ $(m < M)$ become neighbors of the agent $\vartheta_{L,k}$, the priority numbers of their destinations have no effect on the assignment switch and the motion control of agent $\vartheta_{L,k}$ according to Rules 1–6. Thus, the decision process of the agent $\vartheta_{L,k}$ for assignment switch and motion control is the same as that of agent

TABLE I
PARAMETER DESIGN PROCEDURE I

Order	Parameter	Condition
1)	U δ μ	$U > 0$; $\delta > 0$; $0 < \mu \leqslant 0.5$;
2)	R_{STOP} d_0	$R_{\text{STOP}} = \delta + U$; $d_0 > \delta$;
3)	R_{ATTR} R_{SWITCH} p_0	$R_{\text{ATTR}} \geqslant R_{\text{STOP}} + \frac{R_{\text{STOP}}^2}{U}$; $R_{\text{SWITCH}} \geqslant 3R_{\text{STOP}}$; $\|p_{i,0} - p_{j,0}\| \geqslant d_0 \ \forall i, j \in \mathcal{V},\ i \neq j$;
4)	R_{C} d^*	$R_{\text{C}} \geqslant R_{\text{SWITCH}}$; $d^* \geqslant 2R_{\text{ATTR}}$;
5)	Q	$\|q_m - q_n\| \geqslant d^* \ \forall m, n \in \mathcal{V},\ m \neq n$;
6)	θ_0	$\theta_{i,0} \in \mathcal{V},\ \theta_{i,0} \neq \theta_{j,0} \ \forall i, j \in \mathcal{V},\ i \neq j$;
7)	ϕ_0	$\phi_{i,0} = q_{\theta_{i,0}} \ \forall i \in \mathcal{V}$.

$\vartheta_{1,t}$ $(t < k)$ mentioned previously. Hence, $\|q_L - p_{\vartheta_{L,k},k}\|$ will asymptotically decrease to be smaller than R_{ATTR} as k increases to infinity.

In sum, it is shown that $p_{\vartheta_{m,k},k}$ asymptotically converges into $A_{\vartheta_{m,k},k}$ for all $m \in \mathcal{V}$, and thus, $\lim_{k\to+\infty} p_{\vartheta_{m,k},k} = q_m$, or $\lim_{k\to+\infty} p_{i,k} = \phi_{i,k}$ by letting $i = \vartheta_{m,k}$. ■

E. Extension and Parameter Design

Theorem 1 only shows $\|p_{i,k} - p_{j,k}\| > 0$ for collision avoidance. This is not safe enough for practical applications where the agents cannot be treated as mass points regarding their physical size, localization error, and estimated disturbance, etc. That is, we have to guarantee $\|p_{i,k} - p_{j,k}\| > \delta$ for any desired separation distance $\delta > 0$ due to safety considerations. Therefore, all the parameters have to be designed appropriately for such a purpose.

The general idea is to design R_{STOP} properly such that $\|p_{i,k} - p_{j,k}\| \leqslant R_{\text{STOP}}$ always implies $\|p_{i,k+1} - p_{j,k+1}\| \geqslant \|p_{i,k} - p_{j,k}\|$. In the meantime, we need to design a new motion control rule such that $\|p_{i,k} - p_{j,k}\| \geqslant R_{\text{STOP}}$ always implies $\|p_{i,k+1} - p_{j,k+1}\| > \delta$. Then, we can get $d_0 > \delta$ implies $\|p_{i,k} - p_{j,k}\| > \delta$ for all $k > 0$, $i, j \in \mathcal{V}$, $i \neq j$. The detailed parameter design procedure is given in Table I. Before designing the new motion control rule, define the following sets:

$$\begin{aligned} E_{i,k} = \ & \{j \in \mathcal{N}_{i,k} \,|\, \|p_{j,k} - p_{i,k}\| \leqslant 2R_{\text{STOP}}\} \bigcap \\ & \left(\left\{j \in \mathcal{N}_{i,k} \,|\, (\phi_{i,k} - p_{i,k})^{\text{T}} (p_{j,k} - p_{i,k}) > 0\right\} \bigcup \right. \\ & \left. \left\{j \in \mathcal{N}_{i,k} \,|\, (\phi_{j,k} - p_{j,k})^{\text{T}} (p_{i,k} - p_{j,k}) > 0\right\}\right) \\ E'_{i,k} = \ & \{j \in \mathcal{B}_{i,k} \,|\, \|p_{j,k} - p_{i,k}\| \leqslant R_{\text{STOP}}\}. \end{aligned} \tag{28}$$

$E_{i,k}$ stands for the subset of neighbors that may get closer to the agent i if they keep moving at time k. $E'_{i,k}$ is a subset of the blockers of the agent i. Rule 1 is the same as the statement: Set $v_{i,k} = 0$ if $E'_{i,k} \neq \emptyset$. We only need to add a motion control rule to let the agent i stop moving when $E'_{i,k} = \emptyset$ and $E_{i,k} \neq \emptyset$ hold. Note that $E'_{j,k}$ $(j \in E_{i,k})$ can be computed locally by the agent i if $R_{\text{C}} \geqslant 3R_{\text{STOP}}$.

Rule 7: Set $v_{i,k} = 0$ if $E'_{i,k} = \emptyset$ and there exists an agent $j \in E_{i,k}$ such that any one of the following conditions holds:
1) $E_{i,k} \neq \emptyset$, $\|\phi_{i,k} - p_{i,k}\| < R_{\text{ATTR}} - R_{\text{STOP}}$;
2) $E'_{j,k} = \emptyset$, $\|\phi_{j,k} - p_{j,k}\| \geqslant R_{\text{ATTR}} - R_{\text{STOP}}$, $\theta_{i,k} > \theta_{j,k}$ and ($\|p_{j,k} - p_{i,k}\| \leqslant R_{\text{STOP}}$ or $j \in B_{i,k}$).

Remark 5: If $\min_j \|p_{j,k} - p_{i,k}\| > R_{\text{STOP}}$ is further satisfied in 2) of Rule 7 where j stands for all possible agents satisfying the condition 2), stopping movement for the agent i is a little conservative. In this case, we can modify (17) into $v_{i,k} = \min(\lambda_{i,k}\|\phi_{i,k} - p_{i,k}\|, \min_j \|p_{j,k} - p_{i,k}\| - R_{\text{STOP}})$. With no big efforts, it can be proven that the conclusion of (6) still holds. The details are omitted here due to the limitation of space.

By adding the conditions in Rule 7 for setting $v_{i,k} = 0$ into Line 3 of Algorithm 1 and designing the parameters as in Table I, the following conclusions can be obtained.

Lemma 5: $\delta < \|p_{i,k} - p_{j,k}\| \leqslant R_{\text{STOP}}$ implies $\|p_{i,k+1} - p_{j,k+1}\| > \delta$ for all $k > 0$, $i, j \in \mathcal{V}$, $i \neq j$.

Proof: Based on the aforementioned analysis, we only need to show the conclusion holds for $i \in \mathcal{V}$ and $j \in E_{i,k}$. Moreover, if $E'_{i,k} \neq \emptyset$ and $E'_{j,k} \neq \emptyset$, both of agents i and j will stop moving due to Rule 1 and we directly have $\|p_{i,k} - p_{j,k}\| = \|p_{i,k+1} - p_{j,k+1}\| > \delta$. Thus, we only need to consider the case that one or both of them are not affected by Rule 1, i.e., $E'_{i,k} = \emptyset$ or $E'_{i,k} = \emptyset$.

First, assume that both i and j are not affected by Rule 1, i.e., $E'_{i,k} \cup E'_{j,k} = \emptyset$. Without loss of generality and knowing that i and j are homogeneous, we assume $\theta_{i,k} > \theta_{j,k}$. Note that for $j \in E_{i,k}$, $R_{\text{ATTR}} - R_{\text{STOP}}$ cannot be greater than $\|\phi_{i,k} - p_{i,k}\|$ and $\|\phi_{j,k} - p_{j,k}\|$ simultaneously according to Table I. According to Rule 7, if $\|\phi_{i,k} - p_{i,k}\| < R_{\text{ATTR}} - R_{\text{STOP}}$ holds, then $v_{i,k} = 0$; if $\|\phi_{j,k} - p_{j,k}\| < R_{\text{ATTR}} - R_{\text{STOP}}$ holds, then $v_{j,k} = 0$; if $\|\phi_{i,k} - p_{i,k}\| \geqslant R_{\text{ATTR}} - R_{\text{STOP}}$ and $\|\phi_{j,k} - p_{j,k}\| \geqslant R_{\text{ATTR}} - R_{\text{STOP}}$ hold, then $v_{i,k} = 0$. That is, one agent of i and j must stop moving in this case. Second, if either one of i and j is affected by Rule 1, i.e., $E'_{i,k} = \emptyset$ or $E'_{j,k} = \emptyset$ holds but not simultaneously, then the one affected must stop moving. In sum, Rule 7 makes sure that at least one agent of i and j stops moving. Thus, we only need to discuss the results due to the motion of one agent in the sequel.

Without loss of generality, we assume that the agent i stays still, while the agent j moves, then we have $\|\phi_{j,k} - p_{j,k}\| > R_{\text{ATTR}} - R_{\text{STOP}} \geqslant \frac{R_{\text{STOP}}^2}{U}$ according to Table I. Bearing in mind that $\|p_{i,k} - p_{j,k}\| \leqslant R_{\text{STOP}}$, we get

$$\|\phi_{j,k} - p_{j,k}\| > \frac{\|p_{i,k} - p_{j,k}\|^2}{U}. \tag{29}$$

Meanwhile, it can be inferred that $\phi_{j,k} \in S_{j,i,k}$, otherwise $E'_{j,k} \neq \emptyset$ could be implied, which means that the agent j would stop moving due to Rule 1. Denoting by $\alpha_{i,j,k}$ the angle from vector $p_{i,k} - p_{j,k}$ to vector $\phi_{j,k} - p_{j,k}$, we have

$$\cos \alpha_{i,j,k} \leqslant \frac{\|p_{i,k} - p_{j,k}\|}{2\,\|\phi_{j,k} - p_{j,k}\|} < \frac{U}{2\,\|p_{i,k} - p_{j,k}\|}. \tag{30}$$

According to Algorithm 1 and Rule 6, if the agent j is allowed to move, it will move by a distance of U, i.e., $\|p_{j,k+1} - p_{j,k}\| = U$.

Thus, we can get

$$\begin{aligned}\|p_{i,k+1}-p_{j,k+1}\|^2 &= \|p_{i,k}-p_{j,k+1}\|^2\\ &= \|p_{i,k}-p_{j,k}\|^2+\|p_{j,k+1}-p_{j,k}\|^2\\ &\quad -2\|p_{i,k}-p_{j,k}\|\,\|p_{j,k+1}-p_{j,k}\|\cos\alpha_{i,j,k}\\ &> \|p_{i,k}-p_{j,k}\|^2. \end{aligned} \tag{31}$$

Therefore, $\delta < \|p_{i,k}-p_{j,k}\| \leqslant R_{\text{STOP}}$ implies $\|p_{i,k+1}-p_{j,k+1}\| \geqslant \|p_{i,k}-p_{j,k}\| > \delta$. ■

Lemma 6: $\|p_{i,k}-p_{j,k}\| > R_{\text{STOP}}$ implies $\|p_{i,k+1}-p_{j,k+1}\| > \delta$ for all $k>0$, $i,j\in\mathcal{V}$, $i\neq j$.

Proof: If $\|p_{i,k}-p_{j,k}\| > 2R_{\text{STOP}}$, we have $j\notin E_{i,k}$ and it is straightforward to get $\|p_{i,k+1}-p_{j,k+1}\| \geqslant 2R_{\text{STOP}}-2U > 2\delta$ following Rule 6. For $\|p_{i,k}-p_{j,k}\| \leqslant 2R_{\text{STOP}}$, we only need to consider $j\notin E_{i,k}$ and at least one agent of i and j is allowed to move under Rules 1 and 7. If both of them are allowed to move, we have $\|p_{i,k+1}-p_{j,k+1}\| > R_{\text{STOP}}-2\mu U \geqslant \delta$ according to Rule 6 and Table I. Next, we consider the case that only one of them is allowed to move.

Without loss of generality, we assume that the agent i stays still, while the agent j moves. According to Rule 6, it holds that $\|p_{j,k+1}-p_{j,k}\| \leqslant U$. Then, we have

$$\begin{aligned}\|p_{i,k+1}-p_{j,k+1}\| &= \|p_{i,k}-p_{j,k+1}\|\\ &\geqslant \|p_{i,k}-p_{j,k}\|-\|p_{j,k+1}-p_{j,k}\|\\ &> R_{\text{STOP}}-U=\delta. \end{aligned} \tag{32}$$

Therefore, $\|p_{i,k}-p_{j,k}\| > R_{\text{STOP}}$ implies $\|p_{i,k+1}-p_{j,k+1}\| > \delta$ for all $k>0$, $i,j\in\mathcal{V}$, $i\neq j$. ■

Theorem 2: The desired formation can be achieved asymptotically with $\|p_{i,k}-p_{j,k}\| > \delta$ for all $k>0$, $i,j\in\mathcal{V}$, $i\neq j$.

Proof: Given $\|p_{i,0}-p_{j,0}\| \geqslant d_0 > \delta$ for $i,j\in\mathcal{V}$, $i\neq j$ as shown in Table I, $\|p_{i,k}-p_{j,k}\| > \delta$ holds for all $k>0$, which directly follows the conclusions of (5) and (6).

Next, we prove the convergence to the desired formation. The procedure of proof is similar to that of (1). We also start from the convergence of agent $\vartheta_{1,k}$. Whenever $E_{\vartheta_{1,k},k}\neq\emptyset$, it holds that $\theta_{\vartheta_{1,k},k} < \min_{j\in E_{\vartheta_{1,k},k}}\theta_{j,k}$. Thus, Rule 7 has no effect on the motion of the agent $\vartheta_{1,k}$ until $\|q_1-p_{\vartheta_{1,k},k}\| < R_{\text{ATTR}}-R_{\text{STOP}}$ is satisfied. Following the same procedure of the proof of (1), the convergence of the agent $\vartheta_{1,k}$ into $A_{\vartheta_{1,k},k}$ can be shown.

Assume that $\|q_m-p_{\vartheta_{m,k},k}\| \leqslant R_{\text{ATTR}}$ holds for all $m<L$ at time k, where L is a positive integer subject to $1<L\leqslant N$. At time $t>k$, the agent $\|q_L-p_{\vartheta_{L,t},t}\|$ keeps decreasing until the conditions in Rule 1 or Rule 7 are satisfied with respect to $\vartheta_{m,t}$ for some $m<L$. Thus, we will discuss the results in the following three cases for $p_{\vartheta_{L,t},t}\notin A_{\vartheta_{L,t},t}$.

1) Agent $\vartheta_{L,t}$ stops moving at time t due to Rule 1, i.e., $E'_{\vartheta_{L,t},t}\neq\emptyset$ and $\vartheta_{m,t}\in E_{\vartheta_{L,t},t}$ for some $m<L$.

   Since $p_{\vartheta_{L,t},t}\in A_{\vartheta_{m,t},t}$ leads to an active switch for q_L between $\vartheta_{m,t}$ and $\vartheta_{L,t}$, we only need to consider $p_{\vartheta_{L,t},t}\notin A_{\vartheta_{m,t},t}$. In this case, the agent $\vartheta_{m,t}$ keeps moving until $E'_{\vartheta_{m,t},t}\neq\emptyset$ (according to Rule 1) or one of the following two conditions is satisfied for some $g_1<m$ (according to Rule 7).

   1.a) $\vartheta_{g_1,t}\in E_{\vartheta_{m,t},t}$, $E'_{\vartheta_{m,t},t}=\emptyset$, $\|q_m-p_{\vartheta_{m,t},t}\| < R_{\text{ATTR}}-R_{\text{STOP}}$.

   1.b) $\vartheta_{g_1,t}\in E_{\vartheta_{m,t},t}$, $E'_{\vartheta_{m,t},t}\cup E'_{\vartheta_{g_1,t},t}=\emptyset$, $\|q_{g_1}-p_{\vartheta_{g_1,t},t}\| \geqslant R_{\text{ATTR}}-R_{\text{STOP}}$ and ($\|p_{\vartheta_{g_1,t},t}-p_{\vartheta_{m,t},t}\| \leqslant R_{\text{STOP}}$ or $\vartheta_{g_1,t}\in B_{\vartheta_{m,t},t}$);

   If $E'_{\vartheta_{m,t},t}\neq\emptyset$ holds, then Rule 2 or Rule 3 will be triggered leading to the active switches for q_L, which will make $\|q_L-p_{\vartheta_{L,l},l}\| < \|q_L-p_{\vartheta_{L,t},t}\|$ hold at some time $l>t$ after at most $N+2$ active switches following Lemma 3. If the condition 1.a) holds, then we again have $\|q_m-p_{\vartheta_{L,t},t}\| < R_{\text{ATTR}}$, i.e., $\vartheta_{L,t}\in A_{\vartheta_{m,t},t}$. If the condition 1.b) holds, we will consider the motion of agent $\vartheta_{g_1,t}$. If $\vartheta_{g_1,t}$ stops moving at time t, there must be an agent $\vartheta_{g_2,t}$ subject to $g_2<g_1$, $\vartheta_{g_2,t}\in E_{\vartheta_{g_1,t},t}$, $E'_{\vartheta_{g_2,t},t}=\emptyset$, $\|q_{g_2}-p_{\vartheta_{g_2,t},t}\| \geqslant R_{\text{ATTR}}-R_{\text{STOP}}$. By deduction in this way, we can build a chain of agents originating from $\vartheta_{m,t}$ followed by $\vartheta_{g_1,t}$, $\vartheta_{g_2,t}$, and so on, all of which stop moving at time t. Thus, we can find the agent on this chain with the smallest priority number denoted by g^*. Then, it holds that $E'_{\vartheta_{g^*,t},t}=\emptyset$ and $\|q_{g^*}-p_{\vartheta_{g^*,t},t}\| \geqslant R_{\text{ATTR}}-R_{\text{STOP}}$. In this case, the agent $\vartheta_{g^*,t}$ will keep moving until $E'_{\vartheta_{g^*,l},l}\neq\emptyset$ or $\|q_{g^*}-p_{\vartheta_{g^*,l},l}\| < R_{\text{ATTR}}-R_{\text{STOP}}$ holds at some time $l>t$. Otherwise, we can find one more agent with a priority number smaller than g^*. Since $g^*\geqslant 1$ holds, the chain only contains at most m agents. That means the condition 1.b) can only hold for finite length of time so that agent $\vartheta_{m,t}$ will be able to move.

   Therefore, $\|q_L-p_{\vartheta_{L,l},l}\| < \|q_L-p_{\vartheta_{L,t},t}\|$ will hold at some time $l>t$ in the case 1).

2) Agent $\vartheta_{L,t}$ stops moving at time t due to the condition 1) of Rule 7, i.e., $E'_{\vartheta_{L,t},t}=\emptyset$, $E_{\vartheta_{L,t},t}\neq\emptyset$ and $\|q_L-p_{\vartheta_{L,t},t}\| < R_{\text{ATTR}}-R_{\text{STOP}}$ are satisfied.

   In this case, $p_{\vartheta_{L,t},t}\in A_{\vartheta_{L,t},t}$ is inherently implied.

3) Agent $\vartheta_{L,t}$ stops moving at time t due to the condition 2) of Rule 7, i.e., $\vartheta_{m,t}\in E_{\vartheta_{L,t},t}$, $E'_{\vartheta_{L,t},t}\cup E'_{\vartheta_{m,t},t}=\emptyset$, $\|q_m-p_{\vartheta_{m,t},t}\| \geqslant R_{\text{ATTR}}-R_{\text{STOP}}$ and ($\|p_{\vartheta_{L,t},k}-p_{\vartheta_{m,t},k}\| \leqslant R_{\text{STOP}}$ or $\vartheta_{m,t}\in B_{\vartheta_{L,t},k}$) are satisfied for some $m<L$.

We consider the motion of agent $\vartheta_{m,t}$ and can get the same conclusion as in the case 1.b) that it will not stay still all the time and its motion will finally make $E'_{\vartheta_{m,l},l}\neq\emptyset$ or $\|q_m-p_{\vartheta_{m,l},l}\| < R_{\text{ATTR}}-R_{\text{STOP}}$ hold at some time $l>t$. That means agent $\vartheta_{m,t}$ will be able to move at time l to make $\|q_L-p_{\vartheta_{L,l+1},l+1}\| < \|q_L-p_{\vartheta_{L,t},t}\|$ hold.

By summarizing the cases 1)–3), we can show that $p_{\vartheta_{L,k},k}$ asymptotically converges into $A_{\vartheta_{L,k},k}$. By deduction, it holds that $\lim_{m\to+\infty}\|q_L-p_{\vartheta_{m,k},k}\| < R_{\text{ATTR}}$ for all $m\in\mathcal{V}$. If $\vartheta_{m,k}\in A_{\vartheta_{m,k},k}$ is satisfied for all $m\in\mathcal{V}$ (or equivalently, $p_{i,k}\in A_{i,k}$ is satisfied for all $i\in\mathcal{V}$) at time k, all the agents will not be affected by Rule 1 due to $A_{i,k}\cap A_{j,k}=\emptyset$ ($i,j\in\mathcal{V}$, $i\neq j$). Then, the agent $\vartheta_{1,k}$ will never stop moving due to Rule 7, because it has the smallest priority number. As long as $\|q_1-p_{\vartheta_{1,k},k}\| < R_{\text{ATTR}}-R_{\text{STOP}}$ is satisfied, the agent $\vartheta_{1,k}$ will never affect on the motion of other agents via Rule 7. Therefore, we can prove the convergence of $p_{\vartheta_{m,k},k}$ to q_m one by one sequentially for all $m\in\mathcal{V}$, i.e., $\lim_{k\to+\infty}p_{i,k}=\phi_{i,k}$ for all $i\in\mathcal{V}$. ■

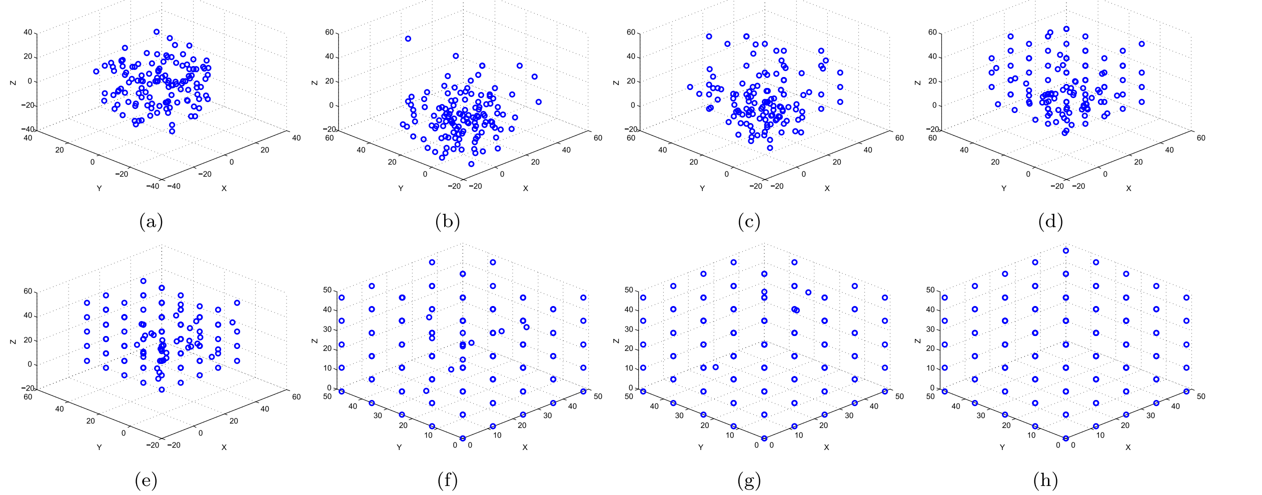


Fig. 4. Snapshots of the agent positions in one simulation. (a) $k = 0$. (b) $k = 15$. (c) $k = 30$. (d) $k = 45$. (e) $k = 60$. (f) $k = 75$. (g) $k = 90$. (h) $k = 105$.

Remark 6: Theorem 2 only considers the distance at the end of each time interval, while in reality, the true distance between agents may be smaller than that. Let t_k be a time instant within the kth time interval $(kT, (k+1)T]$, where T is the length of each time interval. Consider the triangle $\triangle P_{i,kT} P_{i,(k+1)T} P_{j,kT}$ ($j \in E_{i,k}$), we can assume without loss of generality that the agent i moves and the agent j stays still in the kth time interval. It can be found that the minimum value of $\|p_{i,t_k} - p_{j,t_k}\|$ is the shortest distance from $P_{j,kT}$ to the edge ($P_{i,kT}$, $P_{i,(k+1)T}$). Following (30), it holds that $\cos\alpha_{i,j,k} < \frac{U}{R_{\text{STOP}}}$. Then, we have

$$\min_{t_k \in (kT,(k+1)T]} \|p_{i,t_k} - p_{j,t_k}\| \geqslant \|p_{i,kT} - p_{j,kT}\| \sin\alpha_{i,j,k}$$

$$> \delta\sqrt{1 - \cos^2\alpha_{i,j,k}} > \delta\sqrt{1 - \left(\frac{U}{R_{\text{STOP}}}\right)^2} \triangleq \delta^* \tag{33}$$

where δ^* is the lower bound of the separation distance between i and j throughout the kth time interval. Therefore, δ should be designed properly to make δ^* no less than a given threshold in the practical applications.

TABLE II
SIMULATION PARAMETER SETTING

Order	Parameter
1)	$U = 1$; $\delta = 1$; $\mu = 0.25$;
2)	$R_{\text{STOP}} = 2$; $d_0 = 1.1$;
3)	$R_{\text{ATTR}} = 6$; $R_{\text{SWITCH}} = 6$; $R_{\text{C}} = 6$; $p_{i,0}$ $(i \in \mathcal{V})$ are given as random vectors identically independently uniformly distributed over $[-50, 50]^3$ and the case $\|p_{i,0} - q_{j,0}\| \leqslant d_0$ for some $i, j \in \mathcal{V}$, $i \neq j$ is excluded;
4)	$d^* = 12$;
5)	q_m $(m \in \mathcal{V})$ are the points in the set $\{0, d^*, 2d^*, 3d^*, 4d^*\}^3$;
6)	$\theta_{i,k}$=randperm(125), where "randperm(N)" function returns a row vector containing a random permutation of the integers from 1 to 125;
7)	$\phi_{i,0} = q_{\theta_{i,0}}$ $\forall i \in \mathcal{V}$.

IV. Simulation and Experiment

A. Verification by Monte Carlo Simulation

1) Simulation Setup: The simulations are run in MATLAB R2014a [43] and Monte Carlo simulations are implemented. Totally 125 agents are deployed with random initial positions. Monte Carlo simulations are implemented to test the effectiveness of the proposed algorithm under random initial agent positions. The designed parameters used are given in Table II. Snapshots of the agent positions in one simulation are shown in Fig. 4. To test the effectiveness of the proposed algorithm, we define the following two metrics:

$$\text{Dis}_k = \min_{\eta} d_{k,\eta}$$

$$\text{Err}_k = \frac{1}{MN}\sum_{\eta=1}^{M}\sum_{i=1}^{N}\|\phi_{i,k,\eta} - p_{i,k,\eta}\| \tag{34}$$

where M is the total number of Monte Carlo simulations (or experiments), $\eta = 1, 2, \ldots, M$ is the index of simulation, Dis_k indicates the performance of collision avoidance (d_k is defined in (12)), and Err_k indicates the performance of convergence to the desired formation. We implement 1000 Monte Carlo simulations, i.e., $M = 1000$. Dis_k will be compared to δ to verify the correctness of Theorem 2. The convergence time k_{C} is used to evaluate the convergence speed of the proposed method, which is defined as the smallest k such that $\sum_{i=1}^{N}\|\phi_{i,k} - p_{i,k}\| = 0$. It will be compared to the convergence time of the following two typical control methods.

1) All agents move in straight lines to their destinations without assignment switch assuming no collision occurs.
2) The agents move to their destinations one by one in sequence without assignment switch assuming no collision occurs.

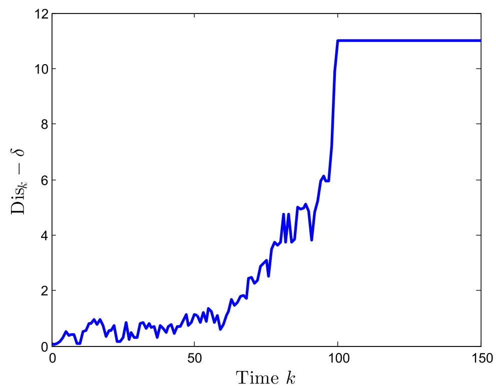


Fig. 5. Results of Dis_k.

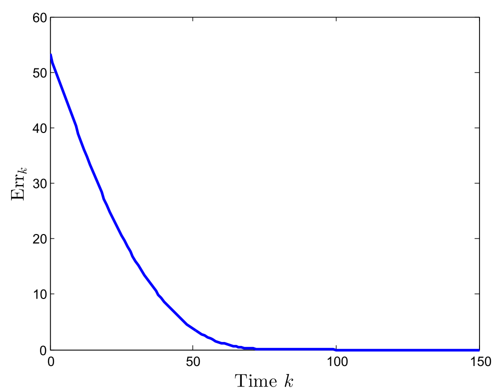


Fig. 6. Results of Err_k.

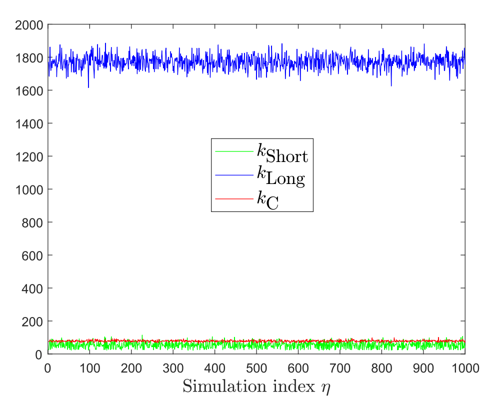
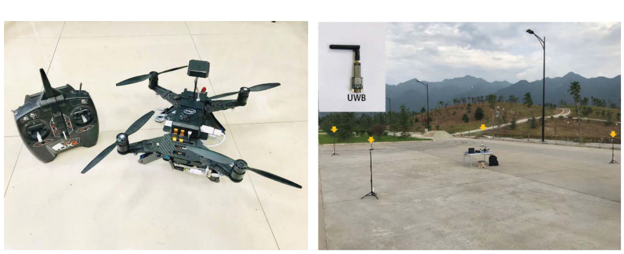


Fig. 7. Results of k_{C}.

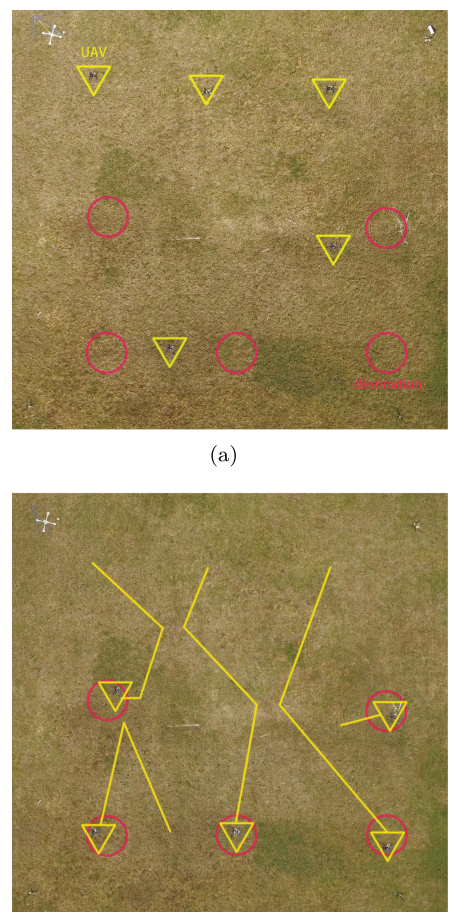


(a) (b)

Fig. 8. Experiment setup. (a) UAV platform. (b) UWB positioning system.

TABLE III
EXPERIMENT PARAMETER SETTING

Order	Parameter
1)	$U = 0.3\mathrm{m/s}$; $\delta = 0.4\mathrm{m}$; $\mu = 0.1$;
2)	$R_{\mathrm{STOP}} = 0.7\mathrm{m}$; $d_0 = 0.5\mathrm{m}$;
3)	$R_{\mathrm{ATTR}} = 2.4\mathrm{m}$; $R_{\mathrm{SWITCH}} = 3\mathrm{m}$; $p_{i,0}$ $(i \in \mathcal{V})$ are given as random vectors identically independently uniformly distributed over $[0, 15]\,\mathrm{m} \times [0, 15]\,\mathrm{m}$ and the case $\lVert p_{i,0} - q_{j,0} \rVert \leqslant d_0$ for some $i, j \in \mathcal{V}$, $i \neq j$ is excluded;
4)	$R_{\mathrm{C}} = 3\mathrm{m}$; $d^* = 5\mathrm{m}$;
5)	$q_1 = [2.5, 12.5]^{\mathrm{T}}$, $q_2 = [7.5, 12.5]^{\mathrm{T}}$, $q_3 = [12.5, 12.5]^{\mathrm{T}}$, $q_4 = [2.5, 7.5]^{\mathrm{T}}$, $q_5 = [12.5, 7.5]^{\mathrm{T}}$;
6)	$\theta_{i,k}$=randperm(5);
7)	$\phi_{i,0} = q_{\theta_{i,0}}$ $\forall i \in \mathcal{V}$.



(a)

(b)

Fig. 9. Actual experimental scenes. (a) Start. (b) End.

The method (1) gives the shortest possible convergence time $k_{\mathrm{Short}} = \max_{i\in\mathcal{V}} \left\lceil \frac{\phi_{i,0}-p_{i,0}}{U} \right\rceil$, while the method (2) gives the longest possible convergence time $k_{\mathrm{Long}} = \Sigma_{i\in\mathcal{V}} \left\lceil \frac{\phi_{i,0}-p_{i,0}}{U} \right\rceil$. This is to evaluate the impact of the assignment switch and the motion control rules on the convergence speed when collision avoidance is involved.

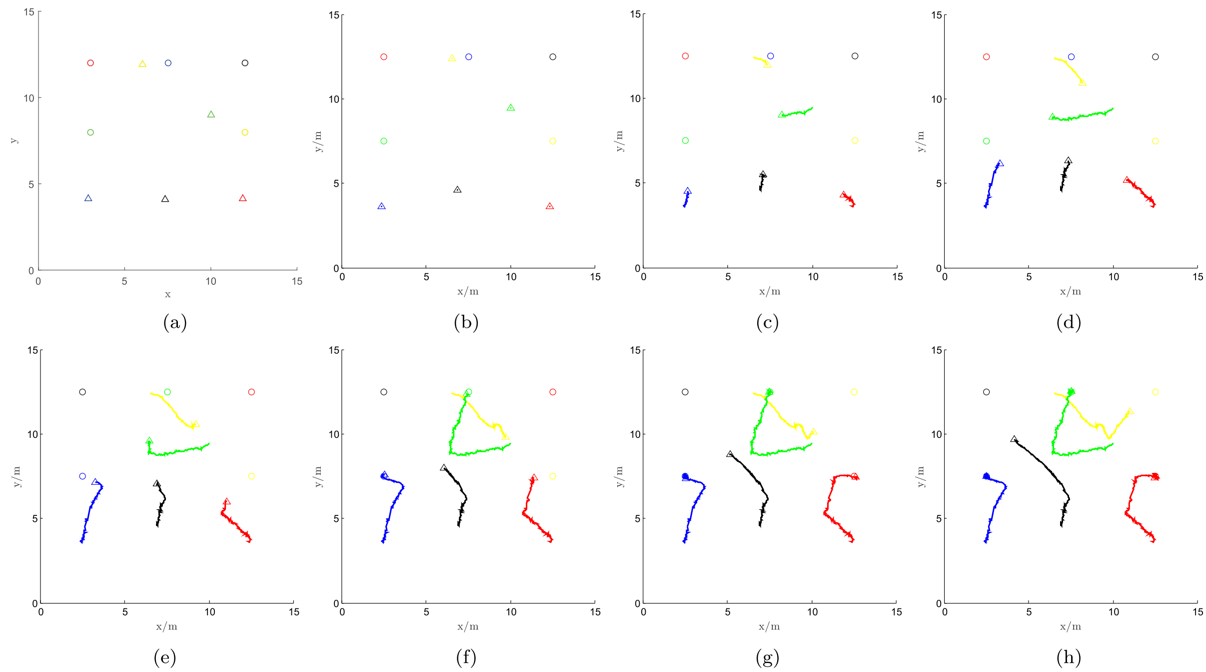


Fig. 10. Snapshots of the recorded trajectories of (a) $k = 0$. (b) $k = 100$. (c) $k = 200$. (d) $k = 300$. (e) $k = 400$. (f) $k = 500$. (g) $k = 600$. (h) $k = 700$.

2) Simulation Results: Fig. 5 shows the results of Dis_k, where $\mathrm{Dis}_k > \delta$ holds for all $k > 0$ in 1000 simulations and $\mathrm{Dis}_k - \delta$ finally converges to $d^* - \delta$. This verifies the effectiveness of the proposed algorithm in collision avoidance. Fig. 6 shows the results of Err_k, which converges to 0 in an exponential pattern, which implies that the convergence is fast enough for practical applications. Fig. 7 shows the results of k_{C}, which is slightly larger than k_{Short}, but much smaller than k_{Long}. This means the proposed method converges nearly as fast as the method (1) even though the agents may stop for a while due to collision avoidance, which is mainly due to the advantage of the assignment switch that helps speed up the convergence.

V. Experiment Demonstration

A. Experiment Setup

Experiments are carried out with randomly selected initial positions of UAVs in the region $[0, 15]$ m $\times$ $[0, 15]$ m. Totally, 20 experiments are implemented, i.e., $M = 20$. In our experiments, five Intel Aero RTF Drones [44] (i.e., $\mathcal{V} = \{1, 2, \ldots, 5\}$) are used as the testing platform [see Fig. 8(a)], flying on the same horizontal plane with 5-m altitude at the speed of no more than 0.5 m/s. The control frequency of the flight control system of each drone is 25 Hz, but we count one second as one time interval and update the way point once per second so that the maximum moving distance U in one step is big enough. The position of each drone in the deployed region is obtained by using an ultra-wideband (UWB) wireless localization system [see Fig. 8(b)] [45]. In an outdoor environment, the UWB positioning system can achieve more stable positioning results with higher accuracy than the usual GPS [46]. The maximum positioning error is less than 10 cm if the distance between each receiver and each anchor is kept within 130 m [44]. The updating frequency of the UWB system for providing positions of the UAVs is 50 Hz so that we can track the motion of drones at high frequency. In our experiment, the UWB anchors are fixed on tripods at the four vertices of the rectangular deployment region. Due to the small localization error, $\phi_{i,k}$ is considered as being reached if $\|\phi_{i,k} - p_{i,k}\| \leqslant 0.1$ is satisfied.

To focus on the performance evaluation of the proposed algorithm, the initial positions are selected such that collision will occur if the agents do not switch destinations. Finally, ten experiments (i.e., $M = 10$) with random initial positions were recorded where the collision avoidance is indispensable. The settings of the related parameters are given in Table III.

B. Experiment Results

Fig. 9(b) shows a snapshot of the scene and the trajectories when the drones converged to the destinations. The whole process of experiment has been recorded into a video and put online (see [47]). The video was taken by a drone carrying a downward camera about 20 m above the five testing drones. The sequential snapshots to show the switches of destinations during moving are presented in Fig. 10, where drone trajectories are denoted by solid lines in different colors (blue, black, red, yellow, and green, respectively) and their destinations are denoted by circles with the same corresponding colors, i.e., the trajectory of a drone has the same color as its destination. It can be seen that the colors of some destinations are changed from time to time. Furthermore, the performance indices Dis_k and Err_k computed from the ten experiments with random initial positions as shown in Figs. 11 and 12 illustrate that the proposed algorithm can

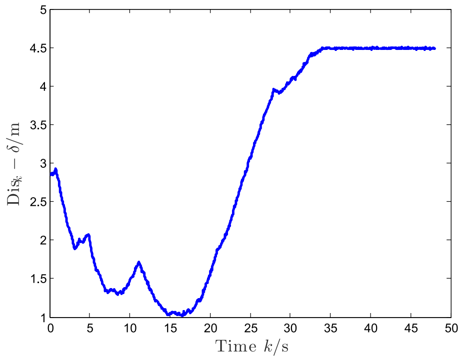


Fig. 11. Results of Dis_k.

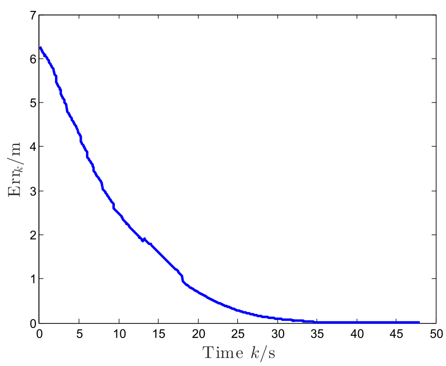


Fig. 12. Results of Err_k.

achieve convergence to the desired formation effectively without collisions.

VI. Conclusion

In this article, a convergent multiagent formation control algorithm was proposed to let a group of homogeneous agents achieve the desired formation under collision avoidance constraint and random initial deployment. The algorithm was fully distributed and only required each agent to know its own position. A detailed parameter design procedure was provided for users to apply the algorithm. The proposed algorithm was verified by Monte Carlo simulations and the actual outdoor experiments.

In the future work, the proposed algorithm will be modified to be adapted in the formation control of fixed-wing UAVs where the turning radius must be considered for collision avoidance. For the fast moving and dense formations, or the so-called swarms, in the wide outdoor environment, the UWB-based or other type of ground-based localization systems are not applicable, while the GPS is not sufficiently accurate. Therefore, novel methods based on relative distance or range will be developed for the convergent formation control of the swarms.

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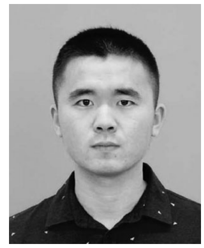

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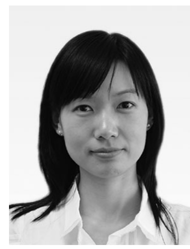

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